

Real Time Research Report (RTRP)

on

AMAZON CLONE

Submitted in partial fulfilment of the requirement for the award of degree

Bachelor of Technology

in

Computer Science and Engineering

by

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Department of Computer Science and Engineering

JOGINPALLY B.R. ENGINEERING COLLEGE
(UGC AUTONOMOUS)

Accredited by NAAC with 'A+' Grade, Recognized under Sec. 2(f) of UGC Act. 1956

Approved by AICTE, Affiliated to JNTUH, Hyderabad and ISO 9001:2015 Certified

Bhaskar Nagar, Yenkapally, Moinabad (Mandal)

R.R. (Dist)-500075. T.S., India

A.Y. 2024-25

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CERTIFICATE

The seminar entitled **AMAZONE CLONE** that is been submitted by **ANGULURI HARI CHARAN-23J21A0506, DANAPALLI SRINIVAS- 23J21A05032, EGYARAPU MADHAN-23J21A0539, GADDI KOPULA VENKATESH- 23J21A0544**, in partial fulfilment of the award of Bachelor of Technology in Computer Science and Engineering to Jawaharlal Nehru Technological University Hyderabad. It is record of bonafide work carried out under our guidance and supervision. In my opinion, this report is of standard required for the degree of Bachelor of Technology.

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DECLARATION

We hereby declare that the Project entitled “**AMAZON CLONE**”, submitted towards the completion of a Real-time Research Project Work in the II-II B.Tech., CSE at Joginpally B.R. Engineering College, Hyderabad, is an authentic record of our original work.

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ABSTRACT

This project is a comprehensive attempt to replicate the core functionalities of Amazon, one of the world's most advanced and widely used e-commerce platforms. The clone simulates major features such as secure user authentication, dynamic product listings, advanced search and filtering, real-time cart management, and order processing. It aims to provide a practical understanding of full-stack web development by combining frontend technologies like HTML, CSS, and JavaScript (or frameworks like React) with backend solutions using Node.js/Express and MongoDB or MySQL for data management. Through this project, students and aspiring developers gain hands-on experience in building scalable and modular web applications while learning critical aspects such as API integration, database design, security implementation, and user interface development. The purpose is not to match Amazon's complexity but to provide an educational platform that bridges the gap between theoretical learning and real-world application development. The final outcome is a functional e-commerce prototype that serves as a strong foundation for further learning or enhancement into a fully deployable system.

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LIST OF ABBREVIATIONS

- UI: User Interface
- DBMS: Database Management System
- SQL: Structured Query Language
- HTML: HyperText Markup Language
- CSS: Cascading Style Sheets
- JS: JavaScript

1. INTRODUCTION

1.1 Objective

The main objective of this project is to design and develop a functional Amazon Clone that replicates the essential features of the original Amazon e-commerce platform. The application will provide features such as user authentication, product listing, search functionality, cart operations, and order management. The project aims to enhance the understanding of full-stack web development by implementing both client-side and server-side logic, offering students and developers a practical learning experience in building scalable web applications.



FIG:1.1

1.2 Scope and Challenges

Scope:

The scope of this project includes building a responsive and user-friendly interface that allows users to register, log in securely, browse through product categories, search for specific items, add products to their cart, and complete orders. The system also maintains user sessions and order history, providing a basic yet functional representation of a real e-commerce platform.

Challenges:

Some of the technical challenges include implementing real-time cart updates, ensuring data consistency between the front-end and back-end, designing an efficient and normalized database structure, handling secure authentication and session management, and developing a fully responsive UI that performs well across different devices. Additionally, integrating a search feature and ensuring scalability in the codebase presents additional complexity.

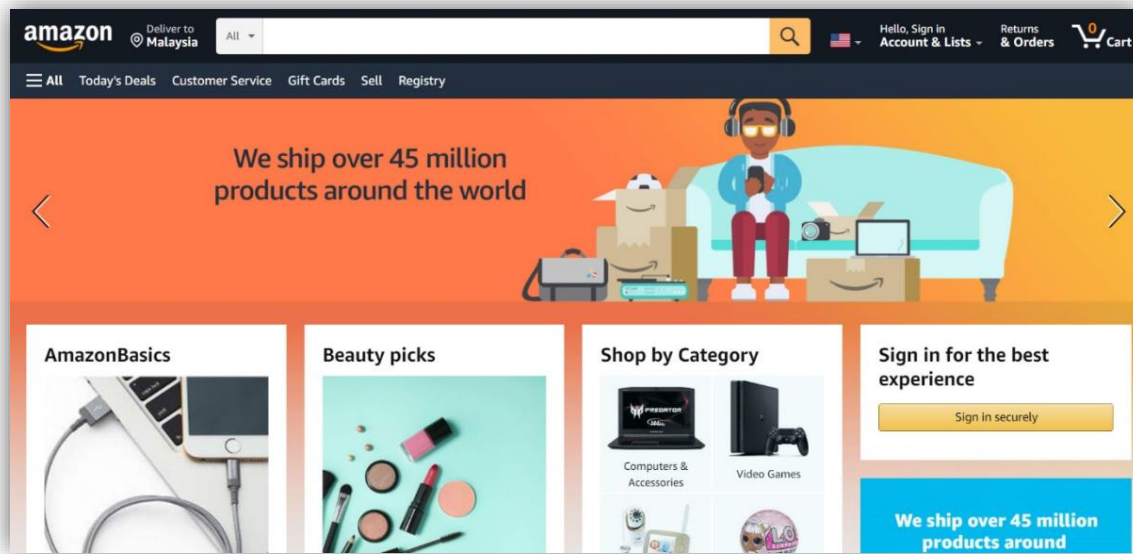


FIG:1.2

1.3 Problem Analysis

Despite the abundance of theoretical resources available through online courses, textbooks, and video tutorials, a significant challenge still exists for learners and aspiring developers when it comes to applying these concepts in a real-world setting. Theoretical knowledge often lacks the contextual depth required to handle full-scale, practical projects, especially in full-stack web development. Most beginner-level projects do not expose students to the complexity of integrating multiple technologies across the frontend, backend, and database layers. Key aspects such as managing user sessions, building responsive and dynamic UIs, implementing authentication and authorization, handling secure payment workflows, or deploying applications on cloud platforms are rarely covered in isolation, let alone together. This disconnect leads to a lack of confidence and competence in solving real industry problems. The Amazon Clone project is designed to bridge this gap by simulating a real e-commerce application. It compels the learner to move beyond static websites and engage in building scalable web applications that interact with databases, offer dynamic functionality through RESTful APIs, and integrate both frontend and backend seamlessly. Through this hands-on experience, students develop a deeper understanding of CRUD operations, database schema design, state management in the frontend, and even DevOps essentials like version control and deployment pipelines—thereby equipping them with industry-relevant skills and confidence in full-stack development.

1.4 Methodology

The project follows a modular, step-by-step development process based on the Software Development Life Cycle (SDLC). The methodology includes:

- Requirement Analysis: Identifying key features and functionalities to replicate from Amazon.
- Design: Creating wireframes and ER diagrams to plan user flow and database structure.
- Development: Building the frontend and backend in parallel using chosen technologies.
- Integration: Connecting front-end UI with back-end APIs and database logic.
- Testing: Manual testing of user flows such as login, add to cart, and checkout.
- Deployment: Hosting the application on a platform such as Vercel, Render, or GitHub Pages for demonstration.

This structured approach ensures better tracking of progress, fewer errors, and a smoother development process.

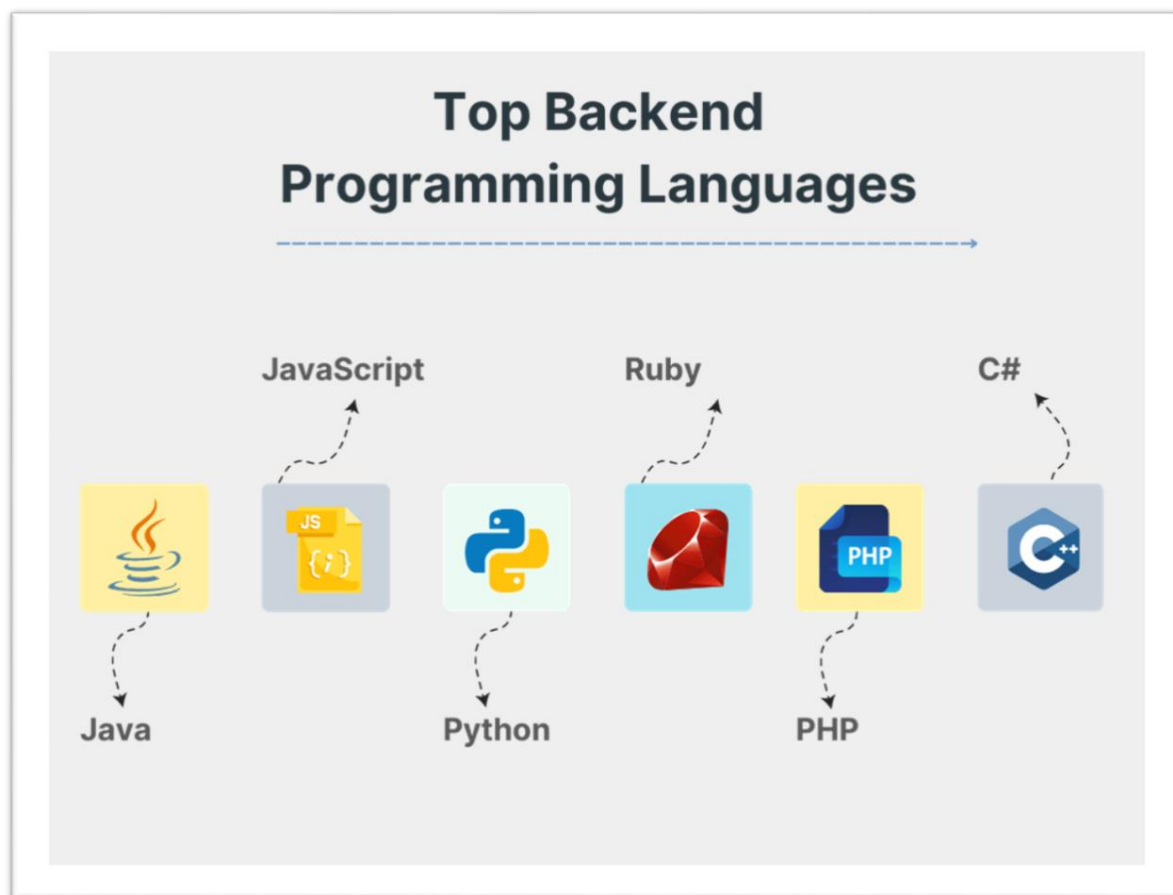


FIG:1.4

1.5 Tools and Technologies Used

The project is developed using a modern full-stack technology suite that allows building scalable and maintainable web applications. The tools and technologies include:

- Frontend:
 - HTML5, CSS3, JavaScript
 - React.js (for component-based UI development)
 - Bootstrap or Tailwind CSS (for responsive design)
- Backend:
 - Node.js (runtime environment)
 - Express.js (web application framework)
- Database:
 - MongoDB (NoSQL database for storing product, user, and order data)
- Other Tools:
 - Postman (for API testing)
 - Git & GitHub (for version control)
 - Visual Studio Code (code editor)
 - Firebase/Render/Vercel (for authentication or deployment, if used)

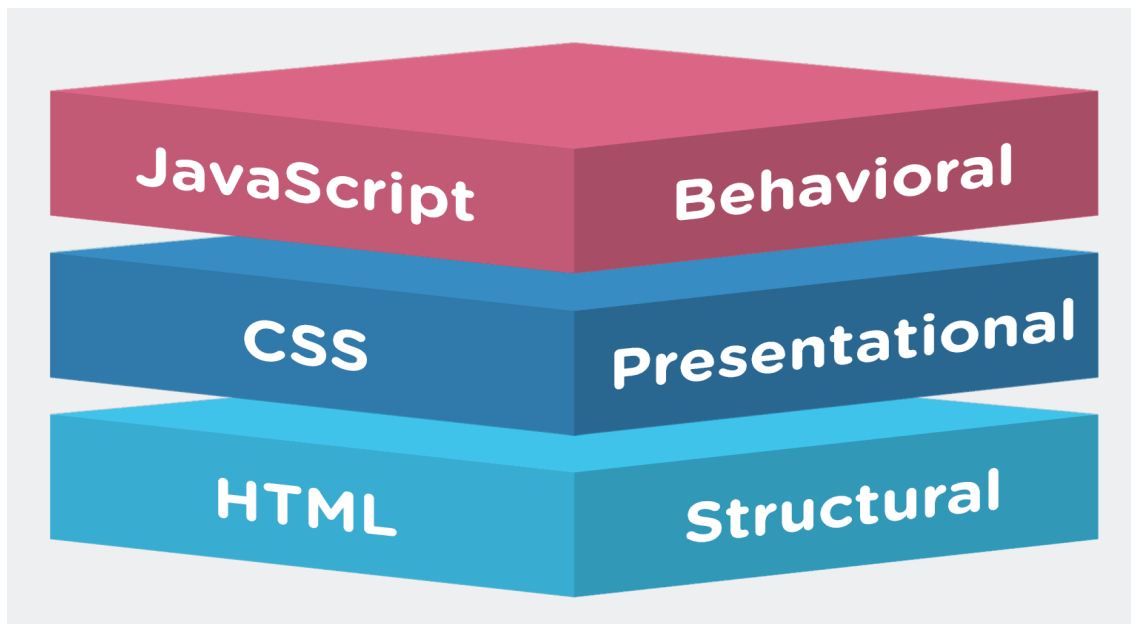


FIG:1.5

This carefully selected technology stack offers a comprehensive learning experience in full-stack web development. By combining essential frontend technologies like HTML5, CSS3, and JavaScript with powerful backend frameworks such as Node.js with Express.js or Django, the stack covers the entire web development lifecycle—from user interface design to server-side logic and database management. Moreover, it maintains a balance between industry relevance and academic simplicity, making it ideal for educational purposes. Students are exposed to real-world development practices including RESTful API creation, state management, user authentication, and CRUD operations, while still working within a framework that is approachable for beginners. This ensures that learners not only understand the theory behind web technologies but also gain hands-on experience in building scalable and maintainable web applications from the ground up.

2. LITERATURE REVIEW

2.1 Existing System Analysis

Amazon is one of the most complex and scalable e-commerce platforms in the world, designed to handle millions of users, thousands of transactions per second, and an extensive inventory of products. Its architecture is built using **microservices**, where each service (like user authentication, product search, cart, payment, etc.) works independently but communicates efficiently through APIs. This provides flexibility and allows each component to scale individually based on demand.

To manage the high volume of traffic and data, Amazon uses **load balancers**, **content delivery networks (CDNs)**, and **distributed databases** such as Amazon DynamoDB. The search engine uses **machine learning algorithms** and **natural language processing (NLP)** to deliver relevant search results and recommendations. Security is ensured through advanced **encryption standards**, **two-factor authentication**, and **secure payment gateways**.

Furthermore, it implements **continuous deployment** and **automated testing** to maintain uptime and reliability.



FIG:2.1

2.2 Areas for Improvement (in the Context of a Clone Project)

While Amazon's system is highly optimized and enterprise-level, it is not feasible or necessary to replicate all of its complexity in a student or beginner-level project. The purpose of a clone is **educational**, so it is important to focus on core functionalities without overcomplicating the architecture. Instead of microservices, a **monolithic architecture** using a single codebase (e.g., MERN stack or Django) is preferred for simplicity. Features such as login, product display, cart handling, and order placement can be implemented using basic CRUD operations and RESTful APIs.

For learning purposes, the use of **third-party APIs** (e.g., payment gateways like Stripe or Razorpay) can be replaced with **mock services** to simulate the functionality. Similarly, advanced algorithms like recommendation engines can be skipped or replaced with hardcoded suggestions. The aim is to make the system **easy to understand, debug, and deploy** so that learners can gain clear insights into how each part of a full-stack application works together.

This simplified clone helps learners to focus on understanding concepts such as **routing, authentication, state management, form handling, and database operations** rather than getting overwhelmed by enterprise-level scalability or infrastructure.

The purpose of a clone is **educational**, so it is important to focus on core functionalities without overcomplicating the architecture. Instead of microservices, a **monolithic architecture** using a single codebase (e.g., MERN stack or Django) is preferred for simplicity. Features such as login, product display, cart handling, and order placement can be implemented using basic CRUD operations and RESTful APIs.

Similarly, advanced algorithms like recommendation engines can be skipped or replaced with hardcoded suggestions. The aim is to make the system **easy to understand, debug, and deploy** so that learners can gain clear insights into how each part of a full-stack application works together.

LITERATURE REVIEW:

Feature	Amazon (Real System)	Clone (Learning System)
Architecture	Microservices	Monolithic (Single codebase)
Database	Distributed (DynamoDB, RDS)	Relational/NoSQL (MySQL, MongoDB)
Search	ML/NLP-based intelligent search	Keyword-based search
Payment	Secure gateway (Stripe, PayPal, etc.)	Simulated checkout
Deployment	Cloud (AWS, multi-region)	Localhost or basic cloud (e.g., Render, Vercel)
Recommendation	Personalized using AI	Hardcoded or category-based

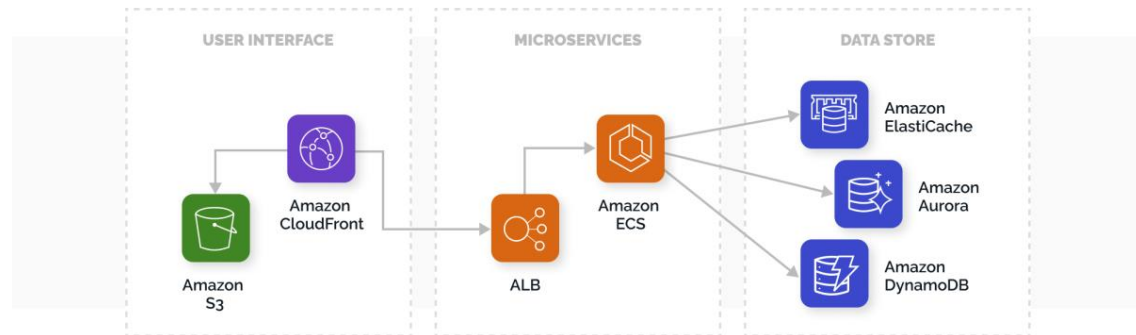


FIG:2.2

3. SYSTEM ANALYSIS

3.1 Hardware Requirements

Processor

The processor is the central hub of all computational activities on a development system. In the Amazon Clone project, which is a full-stack application, the processor plays a critical role in executing and managing multiple simultaneous tasks. The backend logic of the application is typically handled by server-side platforms like Node.js or Django, which, although optimized for performance, require considerable CPU resources when multiple modules and concurrent services are running. These include starting the backend server, connecting to the database, handling HTTP requests/responses, and running authentication or business logic. When integrated with the frontend, the system must also manage live reloading, build tools like Webpack or Vite, and browser rendering processes in real-time. A multi-core CPU (preferably Intel i5/i7 or Ryzen 5/7 series) ensures that each of these services gets adequate processing power without introducing lag, crashes, or throttling. A single-core or low-end processor will struggle to efficiently handle these parallel processes, making development sluggish and debugging tedious. Hence, choosing a good processor not only enhances performance but also significantly improves productivity during the development and testing cycles of the project.



FIG:3.1.1

RAM

Random Access Memory (RAM) is essential for multitasking and handling the short-term memory needs of active applications. For a project like the Amazon Clone, where a variety of tools and services are used simultaneously, having a sufficient amount of RAM is crucial. During development, students or developers typically have multiple applications open: Visual Studio Code (or another IDE), a browser with multiple tabs for testing UI and referencing documentation, terminal windows running servers, Postman for API testing, and even graphic tools for UI/UX adjustments. In addition, services such as MongoDB (running locally) and backend API servers also consume memory in the background. While the absolute minimum RAM required could be 4 GB, it often leads to slowdowns, system freezes, and a poor development experience when multiple tabs or processes are active. An 8 GB or higher RAM configuration ensures smoother performance, faster code execution, efficient debugging, and a responsive UI. It enables the system to cache frequently used files and services in memory, reducing loading times and allowing developers to seamlessly switch between tasks without performance dips. Therefore, for optimal workflow and developer satisfaction, 8 GB RAM or more is highly recommended.

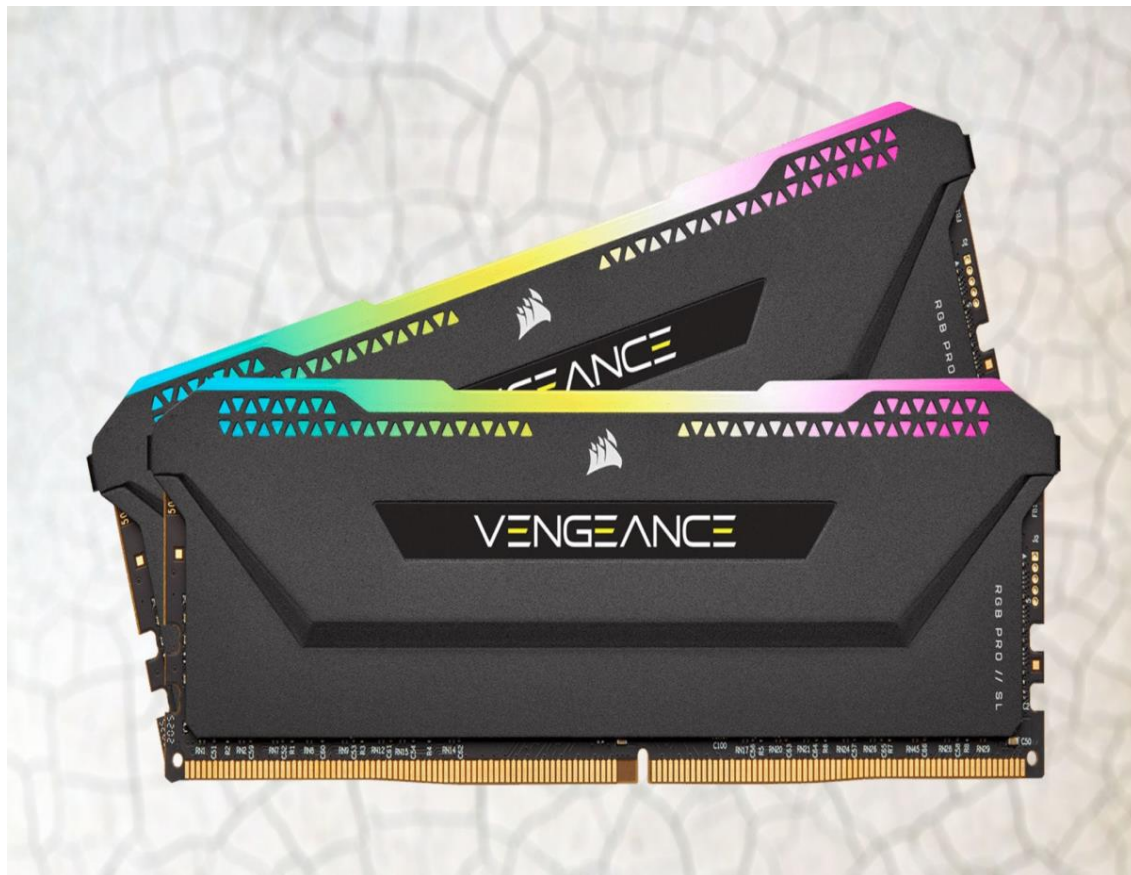


FIG:3.1.2

Storage

Storage is another vital hardware component, particularly for handling the size and complexity of a project like the Amazon Clone. During the course of development, the system needs to install and manage numerous dependencies and libraries using package managers like npm, yarn, or pip. For instance, the `node_modules` directory alone, which contains all the required JavaScript dependencies, can easily exceed hundreds of megabytes. Additionally, the project involves saving product images, user profile pictures, static assets like icons and logos, and even database snapshots. With MongoDB or MySQL databases storing transactional and user data, local storage usage increases further. Beyond that, system logs, cache files, and build outputs also consume significant disk space. Therefore, at least 500 MB of free disk space is required as a bare minimum, although 5–10 GB is more realistic for comfortable development and scalability. Utilizing a **Solid-State Drive (SSD)** instead of a traditional HDD also results in significant performance improvements. SSDs enable faster read/write speeds, reducing the time taken to launch applications, build the project, and load files, which is crucial when frequent builds and hot reloads are part of the development workflow.



FIG:3.1.3

Internet Connection

A reliable and fast internet connection is a necessity for any full-stack web application development process, and it holds especially true for the Amazon Clone project. In the development phase, internet access is required to install packages and libraries using package managers like npm for JavaScript or pip for Python. These packages are hosted on remote repositories and need to be downloaded and updated regularly. Additionally, many frontend components rely on external resources such as Google Fonts, icon libraries like Font Awesome or Heroicons, and CSS frameworks like Bootstrap or Tailwind CSS, which are often fetched from content delivery networks (CDNs). Developers also use GitHub or GitLab to manage version control, collaborate with peers, and push their code to the cloud, which requires consistent connectivity. Moreover, testing and integration of third-party services such as authentication providers (e.g., Firebase or Auth0), payment gateways, or product APIs also necessitate an active internet connection. During deployment, platforms like Vercel, Netlify, or Heroku are used to host the frontend and backend services online. Thus, without a stable internet connection, most stages of the development lifecycle—from coding and testing to deployment and collaboration—come to a halt, making it a non-negotiable hardware requirement.



FIG:3.1.4

3.2 Software Requirements

The Amazon Clone is built using full-stack web development tools. The required software stack includes:

- **Frontend Technologies:**

The frontend is the visual and interactive part of the Amazon Clone that users directly engage with. To develop a responsive, scalable, and visually rich user interface, standard web technologies are used: **HTML5**, **CSS3**, and **JavaScript**. HTML5 forms the semantic backbone of every webpage—it structures content like product listings, buttons, headers, and navigation links. CSS3 enhances the user experience by defining the layout, color schemes, fonts, and responsiveness across different screen sizes using media queries. It plays a vital role in branding and UI aesthetics, including the use of pre-built icon packs and custom styling frameworks like Tailwind CSS or Bootstrap. **JavaScript** powers the dynamic aspects—such as handling form validations, cart updates, dynamic filtering, and UI feedback—making the site feel like a modern, interactive platform. Optionally, frameworks like **React.js** are used to componentize the UI, meaning each element like a product card, search bar, or category filter can be built as an independent, reusable module. React also offers Virtual DOM rendering, which significantly improves performance during frequent UI updates (e.g., when a user filters products or adds them to the cart), enhancing the overall user experience.

- **HTML5:** For content structure
- **CSS3:** For styling and layout
- **JavaScript:** For interactivity and DOM manipulation
- **React.js** (*optional*): For building modular, dynamic UI components

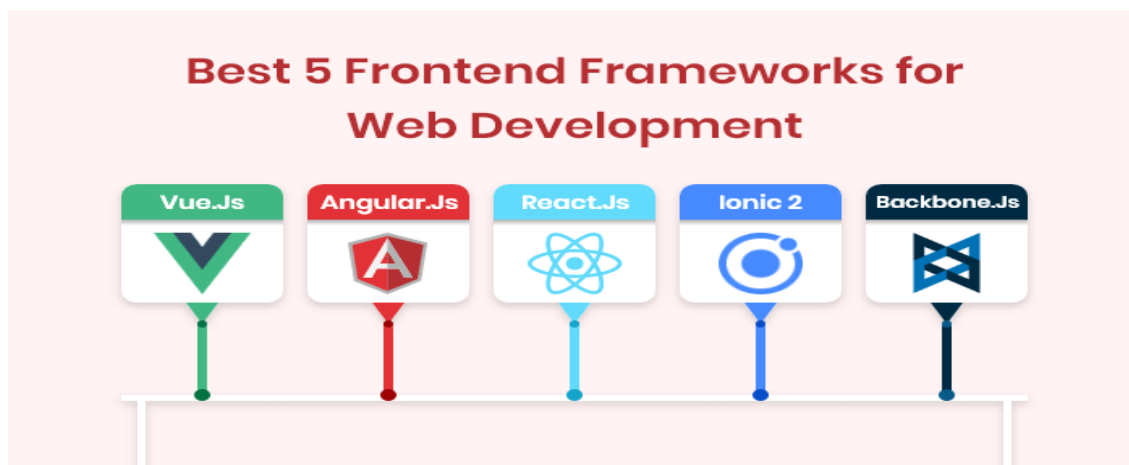


FIG:3.2

- **Backend Technologies:**

The backend of the Amazon Clone handles all logic that the user doesn't see: database operations, server-side routing, authentication, session management, payment processing, and business logic such as inventory control or order history. Two primary backend stacks are considered here—**Node.js with Express.js**, or **Django (Python)**. **Node.js**, a JavaScript runtime environment, is particularly suitable for applications that require real-time operations and can handle a large number of concurrent users due to its non-blocking, event-driven architecture. **Express.js**, a minimalist web framework for Node.js, allows the developer to structure server routes, manage middleware, connect to databases, and handle API requests with simplicity and speed. Alternatively, **Django**, which is based on Python, is ideal for teams or developers who prefer Python's readability and strict MVC (Model-View-Controller) structure. Django comes bundled with powerful features like user authentication, form handling, and an admin dashboard. Both choices offer RESTful API support, making it easy to connect frontend requests to backend data services, ultimately driving a seamless full-stack experience.

- **Node.js + Express.js:** A lightweight, event-driven runtime and web framework for server-side logic
OR
- **Django (Python):** For a structured, MVC-style backend (if Python is preferred)

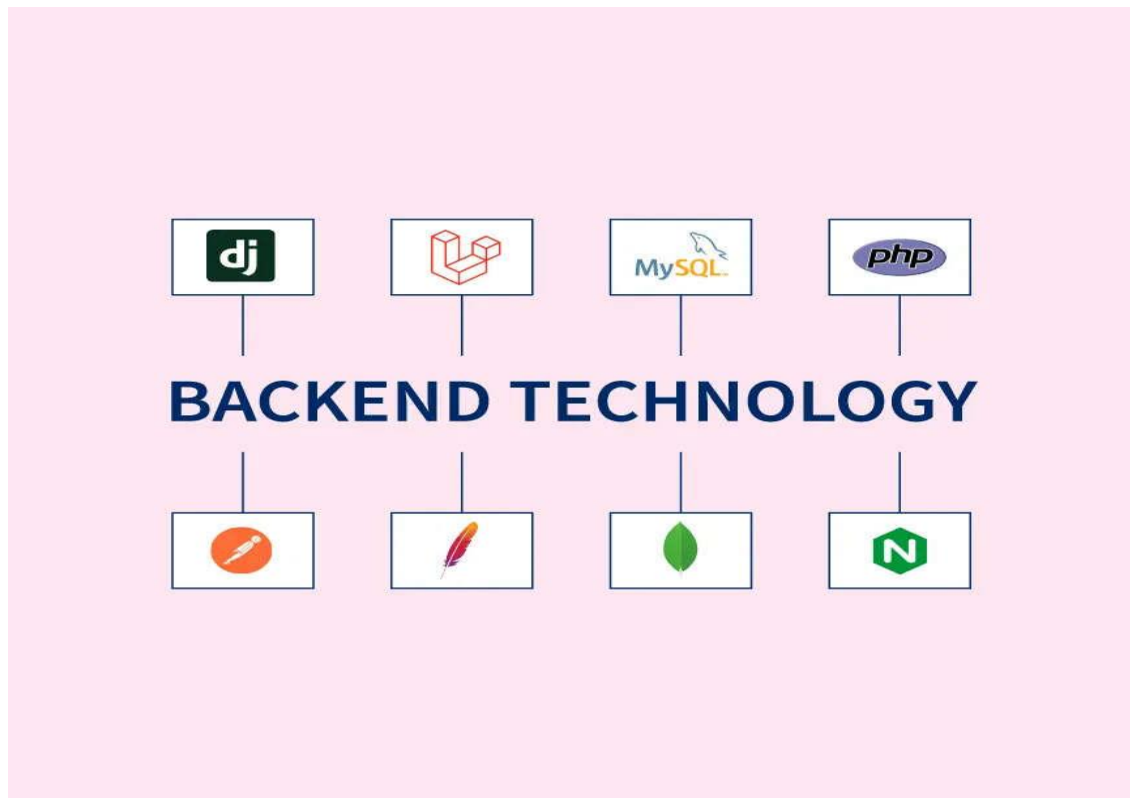


FIG:3.2

- **Database:**

For storing data such as user profiles, product information, cart data, order history, and reviews, a robust database is essential. Two options are primarily considered: MongoDB and MySQL. MongoDB, a NoSQL document-oriented database, is preferred in projects that require flexibility and scale. Its schema-less design allows developers to store JSON-like documents, making it a natural fit when using JavaScript/Node.js across the stack. It supports collections instead of tables, making it easier to store varied product attributes and nested information such as multiple product images or cart details. On the other hand, MySQL is a relational database that enforces structured schemas with defined tables, relationships, and keys. It is a reliable choice for developers using Django, as Django's ORM (Object Relational Mapper) integrates seamlessly with relational databases. MySQL is excellent for applications where data consistency and complex relationships (like managing product inventory across different warehouses) are necessary. Whichever database is chosen, integration with backend code is handled via libraries like Mongoose (for MongoDB) or Django ORM/Sequelize (for MySQL in Node.js).

- **MongoDB:** A NoSQL database, preferred for its flexibility and JSON-like document storage
OR
- **MySQL:** A relational database for structured data with predefined schemas

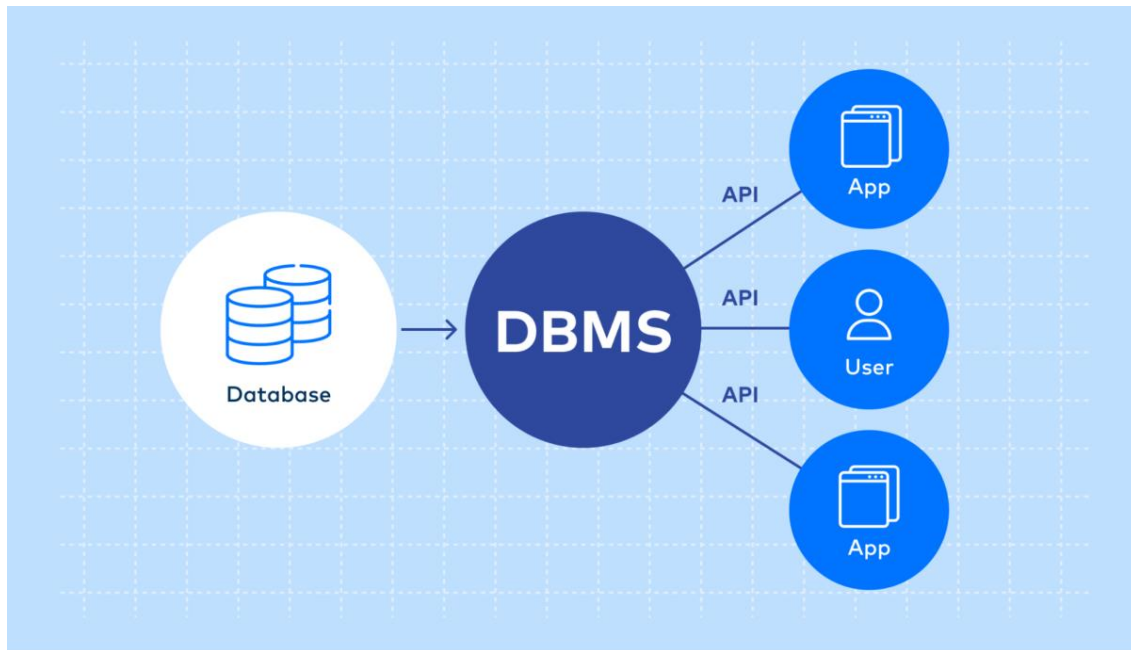


FIG:3.2

- **IDE & Tools:**

In addition to core technologies, several supporting tools and services are used to streamline development. Visual Studio Code (VS Code) is commonly used as the integrated development environment (IDE) because of its robust ecosystem, real-time code linting, terminal integration, and Git support. Node Package Manager (npm) or Yarn is used for installing and managing JavaScript libraries, while pip is used for Python packages in Django-based setups. Postman is a vital tool for testing REST APIs during development. For version control and collaboration, Git and platforms like GitHub or GitLab are employed. These tools allow developers to track changes, manage branches, and work in teams efficiently. For UI/UX design consistency, icon libraries like Font Awesome or Heroicons and color palette generators are also utilized. Additionally, web browsers such as Chrome or Firefox with developer tools assist in real-time debugging, performance checks, and layout validations. This comprehensive software stack ensures that the project is developed in a professional, modular, and scalable manner, suitable for both local deployment and future upgrades like cloud hosting.

1. Visual Studio Code (VS Code)

- Use: Writing and editing your code
- Why it's useful:
 - Supports many languages (HTML, CSS, JS, etc.)
 - Intelligent code suggestions (IntelliSense)
 - Built-in terminal for running commands
 - Extensions for debugging, Git, linting, etc.
- Example: You write your backend logic in Node.js or frontend in React using VS Code.

2. Git

- Use: Version control system — tracks changes in your code
- Why it's useful:
 - Allows you to go back to earlier versions if something breaks
 - Lets you collaborate with others (merge code changes)
 - Works with GitHub for storing your project online
- Example: You create a Git repo for your Amazon Clone and push changes regularly.

3. Postman

- Use: Testing and debugging APIs (backend services)
- Why it's useful:
 - You can simulate HTTP requests (GET, POST, PUT, DELETE)
 - Check if API endpoints are working correctly
 - Send data as JSON to test backend responses
- Example: Before linking the frontend, you test your /login API using Postman.

4. Browser (Chrome or Firefox)

- Use: Testing and debugging the frontend (UI/UX)
- Why it's useful:
 - Check how your website looks and behaves
 - Use DevTools (F12) to inspect HTML/CSS, debug JavaScript
 - Test responsiveness on different screen sizes
- Example: View and debug your Amazon Clone's product listing page in Chrome.

5. npm / yarn

- Use: Managing project dependencies and libraries
- Why it's useful:
 - Install packages (like express, mongoose, bcrypt, react, etc.)
 - Run scripts (like npm start or yarn build)
 - Automatically handle package versions
- Example: You run npm install to set up your backend server dependencies for your clone.



Visual Studio Code

FIG:3.2

This stack is sufficient for building a complete e-commerce application, from front-end design to back-end logic and database connectivity.

3.3 Data Collection Process

The data required for the application (such as user accounts, product details, and orders) is simulated using:

In developing the Amazon Clone project, the data collection process plays a vital role in simulating a realistic e-commerce environment while maintaining full control for testing and educational purposes. Since this is a clone application created for learning, the dataset is not sourced from a live database but is instead generated using multiple structured and flexible approaches. One primary method is **dummy data seeding**, where sample product catalogs, user profiles, and test order entries are manually crafted and inserted into the system. These entries include typical product information such as name, price, category, rating, and availability status, along with basic user attributes like username, email, and encrypted passwords. Another method involves using **predefined JSON files or mock APIs** that contain structured data resembling real-world entities. These files are either imported directly into the database during project setup or used as endpoints to mimic external API responses, which helps simulate asynchronous data handling in the frontend. Additionally, **dynamic data entry through form inputs** is enabled—both regular users and admin users can create or modify records such as product listings or account information during runtime. This adds interactivity and helps validate CRUD functionalities within the system. Furthermore, **image assets for products** are sourced from publicly available image repositories, such as Unsplash or Lorem Picsum, or replaced by placeholder graphics during development. These assets are integrated into product cards to replicate a realistic product gallery layout. Collectively, these methods ensure the availability of a consistent, customizable dataset that supports frontend rendering, backend processing, and user interaction—creating a complete end-to-end simulation of data flow in a real-world e-commerce platform.

- **Dummy Data Seeding:** Manually created product catalogs, user profiles, and test orders
- **JSON Files or APIs:** Predefined product or user data in JSON format to populate the database during setup
- **Form Inputs:** Registered users and admin can add or edit product listings during runtime
- **Image Assets:** Product images are sourced from open sources or placeholders

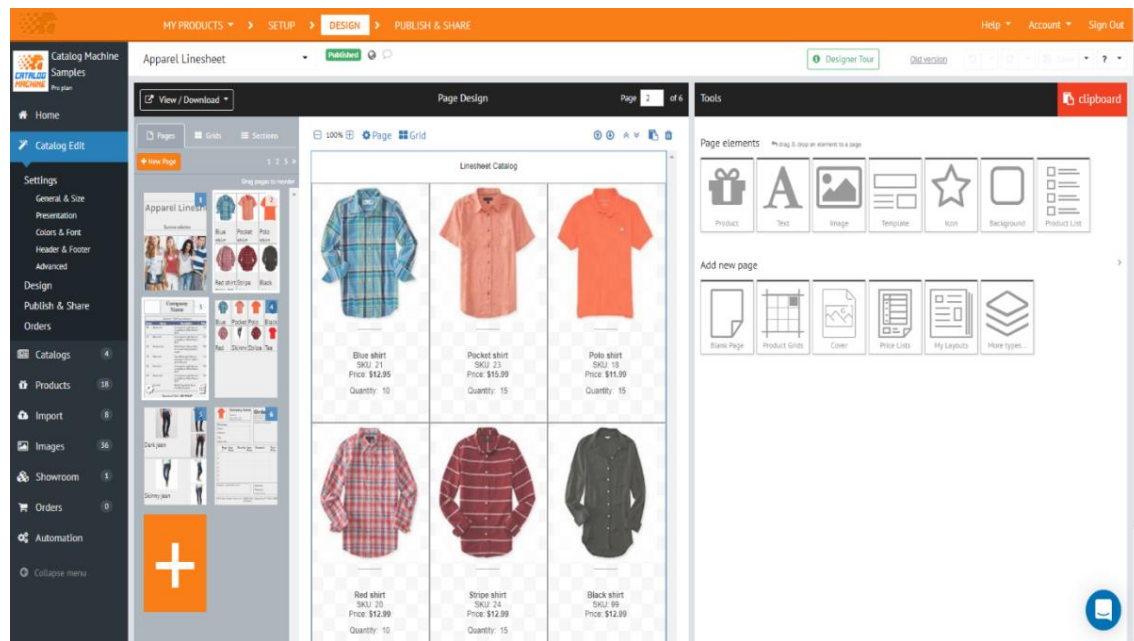


FIG:3.3

This approach ensures a controlled and customizable data set for both testing and demonstration purposes.

3.4 Beneficiaries

This project benefits multiple user groups:

The Amazon Clone project serves as a valuable resource for a diverse group of stakeholders, each benefiting from its educational and practical significance. Students, particularly those pursuing computer science or web development courses, gain hands-on experience in full-stack development through this project. It allows them to apply theoretical concepts in real-world scenarios, reinforcing their understanding of frontend-backend integration, database operations, and UI/UX design. Aspiring developers and fresh graduates can use the project as a strong addition to their professional portfolios, demonstrating their ability to build scalable, real-time applications using modern technologies. This can significantly enhance their prospects during job interviews or internship applications. Furthermore, startups and entrepreneurs can use the clone as a foundational structure for developing Minimum Viable Products (MVPs) for small-scale e-commerce platforms. Instead of building an application from scratch, they can customize and extend the clone to suit specific business requirements, reducing development time and cost. Educators and instructors also benefit from this project, as it provides a complete and practical case study to teach web development, project structuring, and deployment. Additionally, the collaborative nature of this type of project encourages group participation, code sharing, and role-based development, making it ideal for classroom or academic group assignments. Overall, the Amazon Clone project not only facilitates technical growth but also promotes project management, team collaboration, and problem-solving skills in a controlled yet realistic development environment.

- **Students:** Enables learning of full-stack development and hands-on application building
- **Aspiring Developers:** Acts as a portfolio project to showcase skills in interviews or job applications
- **Startups/Entrepreneurs:** Can serve as a foundation for building MVPs (Minimum Viable Products) for small-scale online stores
- **Educators:** Useful as a teaching aid for web development courses

The clone also encourages collaboration and group project opportunities in academic environments.

3.5 Algorithms Implemented

Several simple but effective algorithms and logic flows are used in the application:

- **User Authentication:**
 - Using **JWT (JSON Web Tokens)** or **Session-Based Authentication**
 - Logic: Validate user credentials → generate token/session → protect private routes
- **Product Filtering/Search:**

- Case-insensitive matching of keywords in titles or descriptions
- Optional category-based or price-range filters
- **Cart Management:**
 - Add/update/remove products with quantity checks
 - Calculate subtotal and display live cart status
- **Order Checkout Process:**
 - Collect delivery details, display cart summary, simulate payment process
 - Save order data and clear cart on successful checkout

Dashboard Quotes Sales Clients Orders Products										
Search By order ID By product name, sku, title... By name (show/hide advanced filters) By order amount From To By coupon By coupon code By date From To By order status show all By order type show all By shipping destination shipping country shipping state shipping carrier Apply filter Reset all filters										
	#	Customer name #	Email #	Purchased on #	Status #	Products in cart #	Total price #	Shipping price #	Shipping via	Invoice & packing slip Order type #
	356	Alex RB	alexandra8@peosamba.com	19-Feb-2016	Items Shipped	1	606.00	0.00	pickup Tracking ID: vvd7 [edit]	view Regular
	355	Alex RB	alexandra8@peosamba.com	19-Feb-2016	Items Shipped	1	786.50	0.00	pickup Tracking ID: Colissimo 9400317441411 http://www.colissimo.fr/portal_colissimo/maire.do?m=10003005&colispart=http://www.colissimo.fr/portal_colissimo/maire.do?m=10003005&colispart=9400317441411 [edit]	view Regular
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FIG:3.5

These are core logic blocks that simulate the typical behavior of an e-commerce site without external dependencies.

3.6 Methodology

The development approach follows the **Agile Software Development Life Cycle (SDLC)**, which allows for iterative progress and continuous feedback:

1. **Planning:** Identify features and define user stories (e.g., login, search, checkout)
2. **Design:** Build wireframes for the UI and create database schemas
3. **Development:** Implement modules in short sprints, often working in parallel (frontend/backend)
4. **Testing:** Perform unit and integration tests to validate features
5. **Deployment:** Host the application using platforms like Vercel, Render, Firebase, or GitHub Pages
6. **Maintenance:** Apply feedback, fix bugs, and update features post-deployment

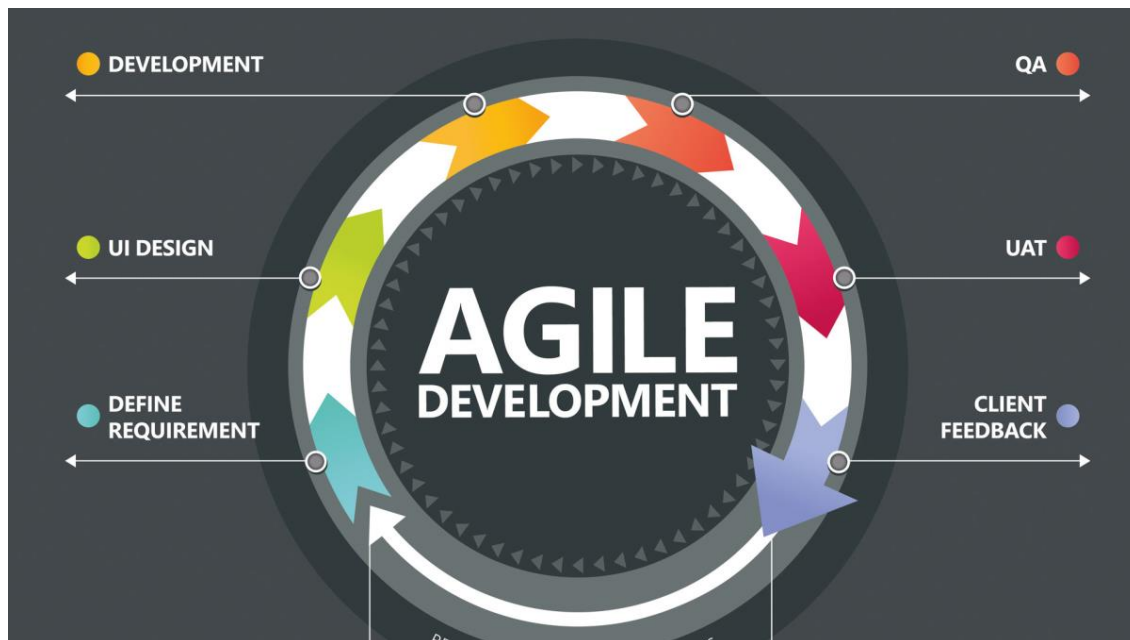


FIG:3.6

Agile methodology promotes flexibility, quicker iterations, and collaboration, making it ideal for student and team-based projects.

4. SYSTEM FEASIBILITY

4.1 Economic Feasibility

This project is **highly economical**, especially for academic and prototype-level implementations. The major reasons are:

- **No Commercial Licensing:** Uses free, open-source technologies (HTML, CSS, JS, Node.js, MongoDB).
- **Hardware Requirements:** Can be developed and tested on standard laptops without dedicated servers.
- **Free Hosting Options:** Deployment can be done via **Render, Vercel, or GitHub Pages** at zero cost.
- **Learning Resources:** Free tutorials and documentation available (e.g., MDN, GeeksforGeeks).
- **Team-Based Development:** Multiple developers can contribute using **Git & GitHub** without any enterprise tools.

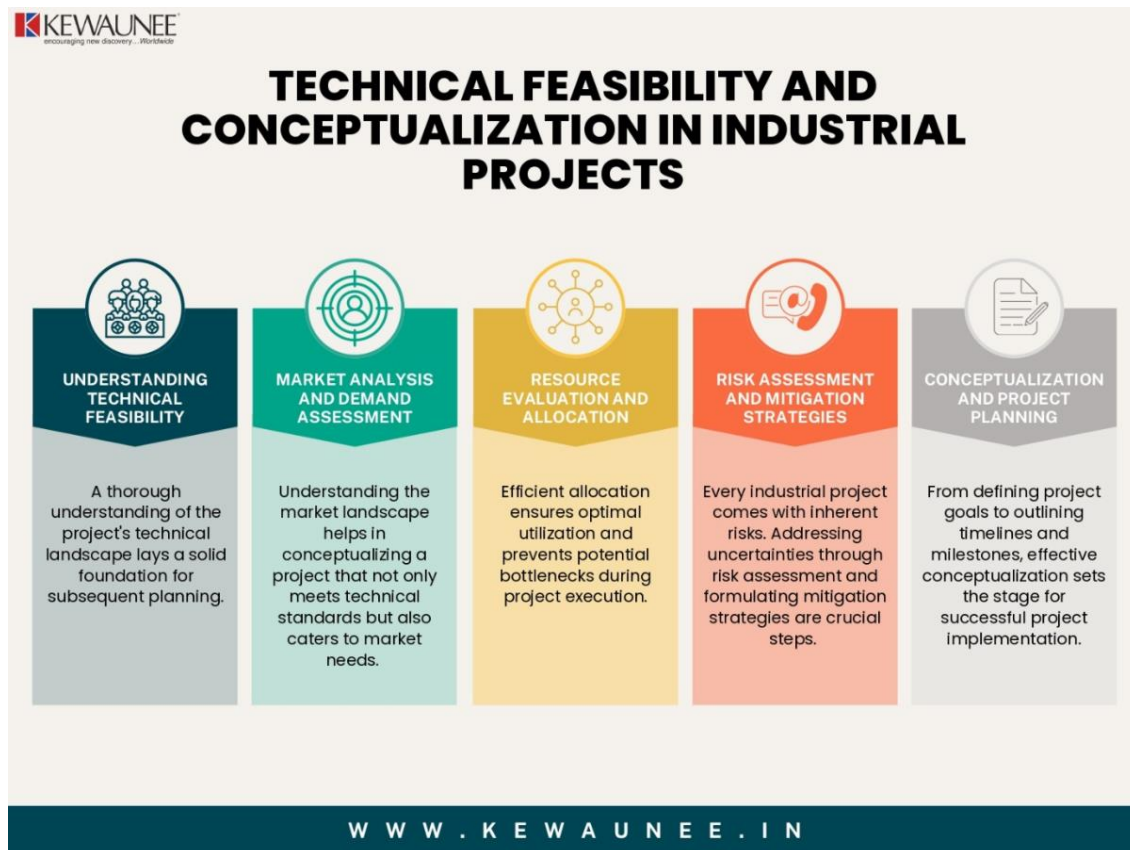


FIG:4.1

Hence, the return on investment (ROI) is high for learning and small-scale demo purposes.

4.2 Technical Feasibility

The system is considered **technically feasible** because it's built using **widely-used web technologies** that are modular, open-source, and easy to integrate. Here's what each part means:

Frontend:

- **HTML (HyperText Markup Language):** This is the structure of your web page — like the skeleton. It defines headings, buttons, inputs, images, etc.
- **CSS (Cascading Style Sheets):** This handles the design and appearance — colors, layout, font size, animations, and responsiveness.
- **JavaScript:** Adds interactivity like dropdowns, form validation, showing/hiding elements, and handling click actions.
- **React (Optional Framework):** A JavaScript library that helps in building complex UIs using components (reusable UI blocks). It's faster, cleaner, and used in many real-world apps.

Backend:

- **Node.js:** A runtime that lets you run JavaScript on the server (not just in browsers). It's fast and supports many users at once because it handles requests *asynchronously* (doesn't block others while processing).
- **Express.js:** A lightweight framework on top of Node.js that simplifies route handling and server logic.

Database Integration:

- **MongoDB:** A NoSQL database that stores data in flexible JSON-like format (documents). Perfect for e-commerce apps where products might have different attributes (e.g., clothes vs electronics)..

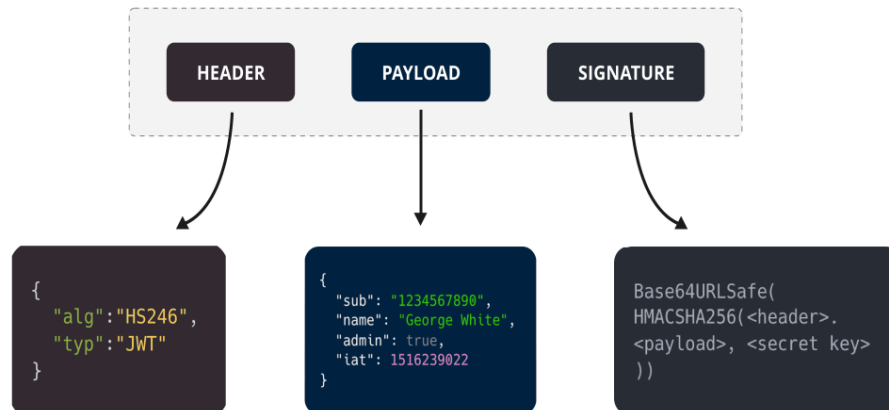


FIG:4.2

Authentication:

- **JWT (JSON Web Token):** A token (like a digital ID) that's issued after login. It's stored on the frontend and sent with every API request to prove the user is authenticated.
- **Session-based authentication:** The server keeps track of logged-in users and assigns a session ID (stored in browser cookies).
- **Passport.js:** A popular middleware to help handle both JWT and session-based authentication.

Structure of a JSON Web Token (JWT)



SuperTokens

FIG:4.2

Security Practices:

- **Data Validation:** Checking user input (e.g., making sure email is valid, passwords aren't empty, etc.).
- **HTTPS (SSL):** Ensures data sent between server and browser is encrypted (secure).
- **Password Hashing:** Using libraries like bcrypt to convert passwords into encrypted formats so they can't be stolen easily.
- **Access Control:** Restricting access to certain features (e.g., only logged-in users can place orders, only admins can add products).



FIG:4.2

API-First Architecture:

- **API (Application Programming Interface):** A set of rules for how the frontend talks to the backend.
- **RESTful API:** Uses standard HTTP methods like:
 - GET (fetch data),
 - POST (send data),
 - PUT (update data),
 - DELETE (remove data).

API-First Architecture: RESTful APIs enable modular backend logic and smooth frontend-backend communication.



FIG:4.2

It is feasible even for intermediate developers and scalable if optimized further.

4.3 Social Feasibility

The system addresses **educational, entrepreneurial, and societal value:**

- **Educational Use:** Helps students and self-learners gain hands-on experience in full-stack development, MVC architecture, authentication, and CRUD operations.
- **Startup Prototyping:** Early-stage e-commerce ideas can use this clone to test business models or demo features.
- **Community Learning:** The clone can be improved collaboratively (open-source), encouraging peer learning and innovation.
- **Job Preparation:** Acts as a portfolio project for full-stack or backend developer roles.

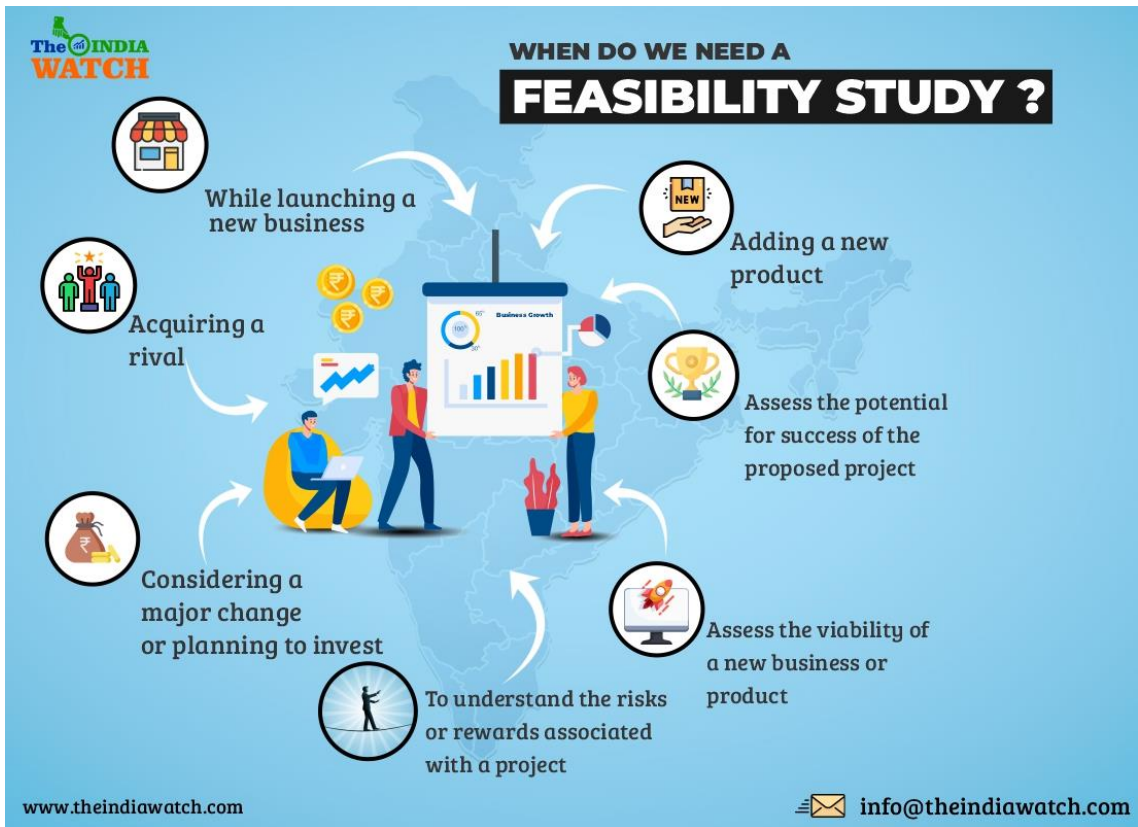


FIG:4.3

5. SYSTEM DESIGN

5.1 ER Diagram

Entities & Relationships:

- **User**
 - Fields: user_id, name, email, password, role
 - One-to-many → Orders
- **Product**
 - Fields: product_id, name, description, category, price, stock
 - Many-to-many → Users (via Cart and Orders)
- **Cart**
 - Fields: cart_id, user_id, product_id, quantity
 - Each user can have multiple cart items
- **Order**
 - Fields: order_id, user_id, product_list, total_amount, status, created_at
 - Tracks what was bought and when

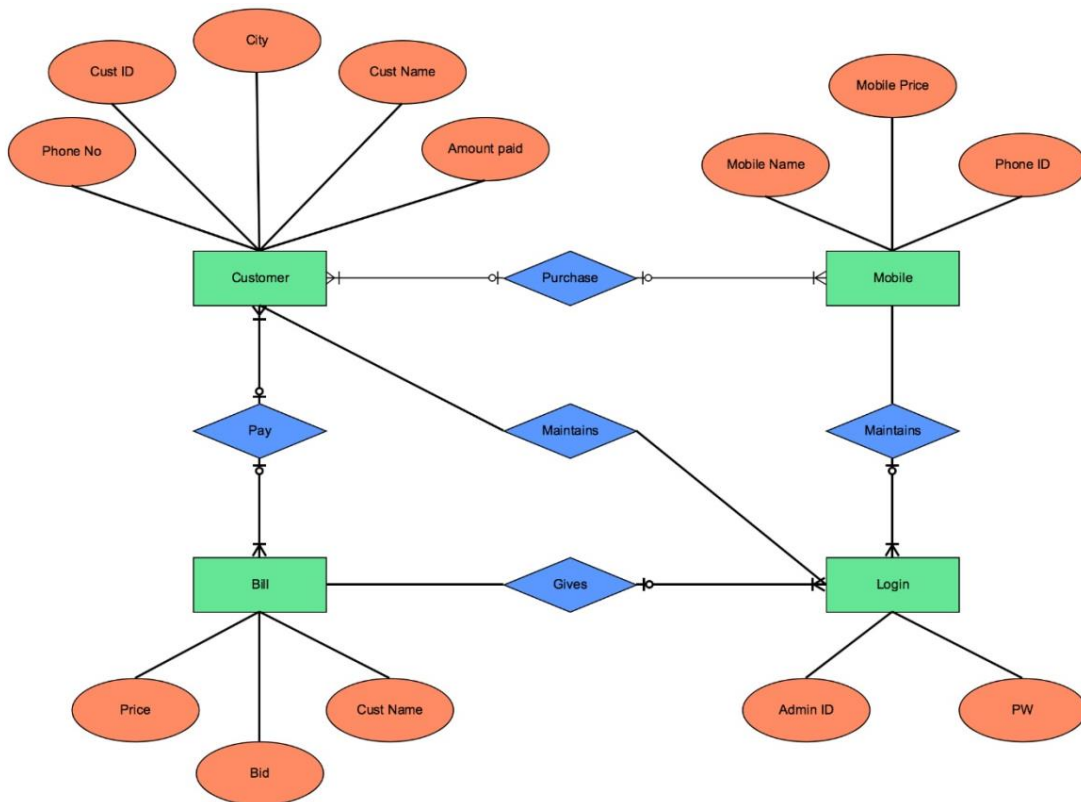


FIG:5.1

5.2 System Architecture

- **Client Layer (Frontend):**
 - Renders UI, fetches API data, handles routing using React Router or similar.
 - Axios/Fetch used to communicate with backend APIs.
- **Server Layer (Backend):**
 - Node.js manages request routing, session control, JWT issuance, CRUD APIs.
 - Middleware (body-parser, cors, auth-checkers) ensure secure communication.
- **Database Layer:**
 - MongoDB stores documents (JSON-like), ideal for variable-length product specs.
 - Uses collections (users, products, orders).
- **REST APIs:** Stateless communication using HTTP verbs (GET, POST, PUT, DELETE).

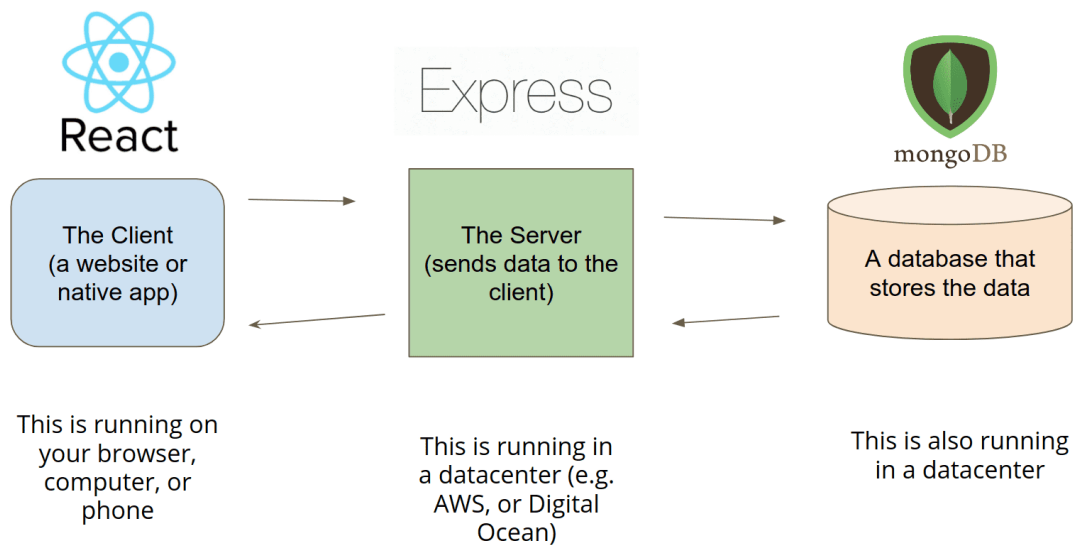


FIG:5.2

6. SYSTEM IMPLEMENTATION

6.1 Modules

6.1.1 User Login

- Register and login forms with field validation (e.g., email format, password length).
- Passwords hashed using bcrypt before saving.
- On login, a **JWT token** is issued and stored in the browser (localStorage or cookies).
- Private routes are protected using middleware that checks for token presence and validity.

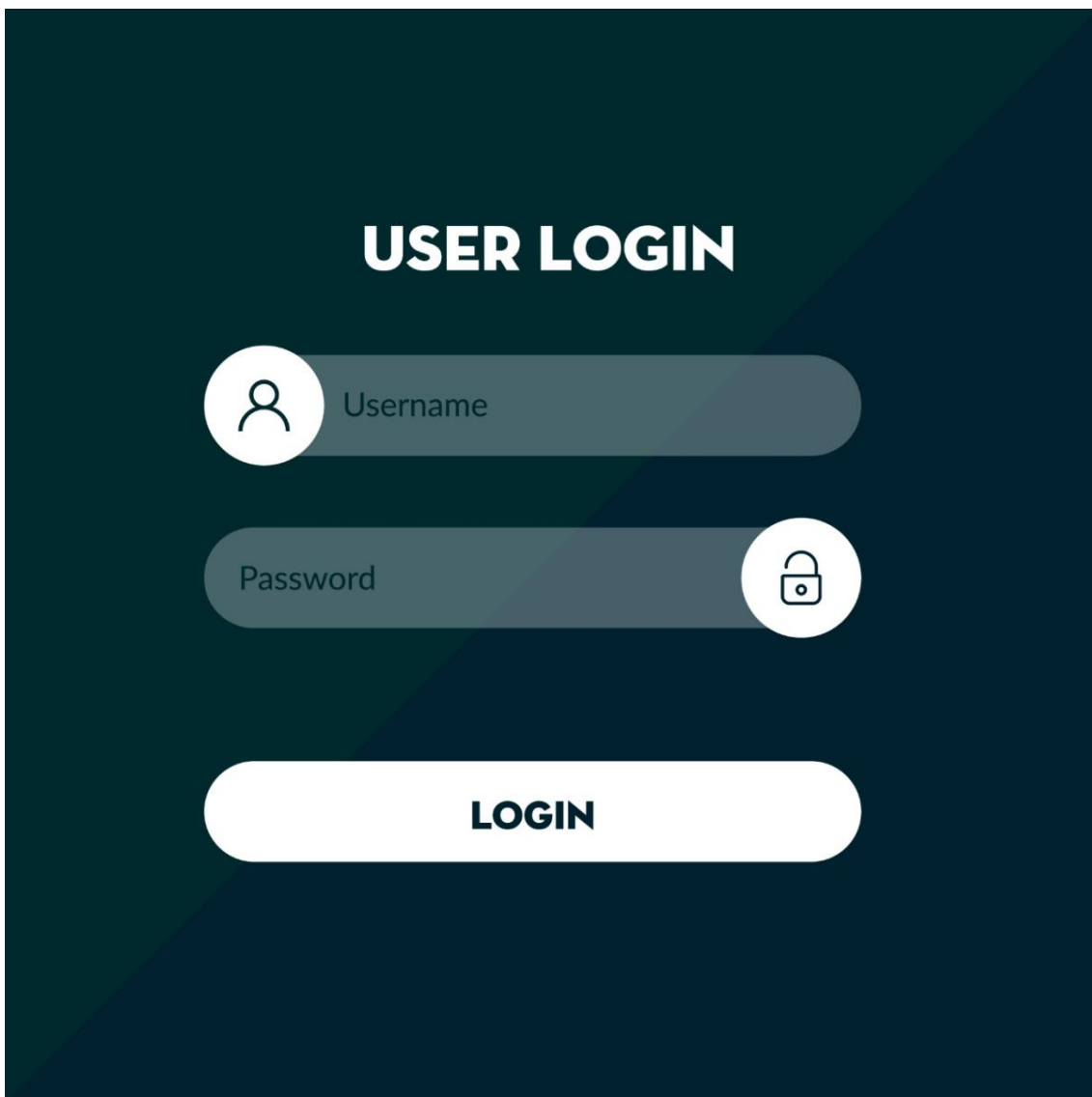
The image shows a user login interface on a dark blue background. At the top, the text "USER LOGIN" is displayed in large, bold, white capital letters. Below this, there are two input fields. The first field is for the "Username", indicated by a person icon in a white circle on the left and the text "Username" inside a light blue rounded rectangle. The second field is for the "Password", indicated by a padlock icon in a white circle on the right and the text "Password" inside a light blue rounded rectangle. Below these fields is a large, white, rounded rectangular button with the word "LOGIN" in bold, dark blue capital letters.

FIG:6.1.1

6.1.2 Product Browsing

- Homepage fetches a list of products from the database via API.
- Features:
 - Filter by category or brand
 - Search bar with keyword match
 - Sort by price/ratings
- Each product is a clickable card showing image, price, stock status, and description.

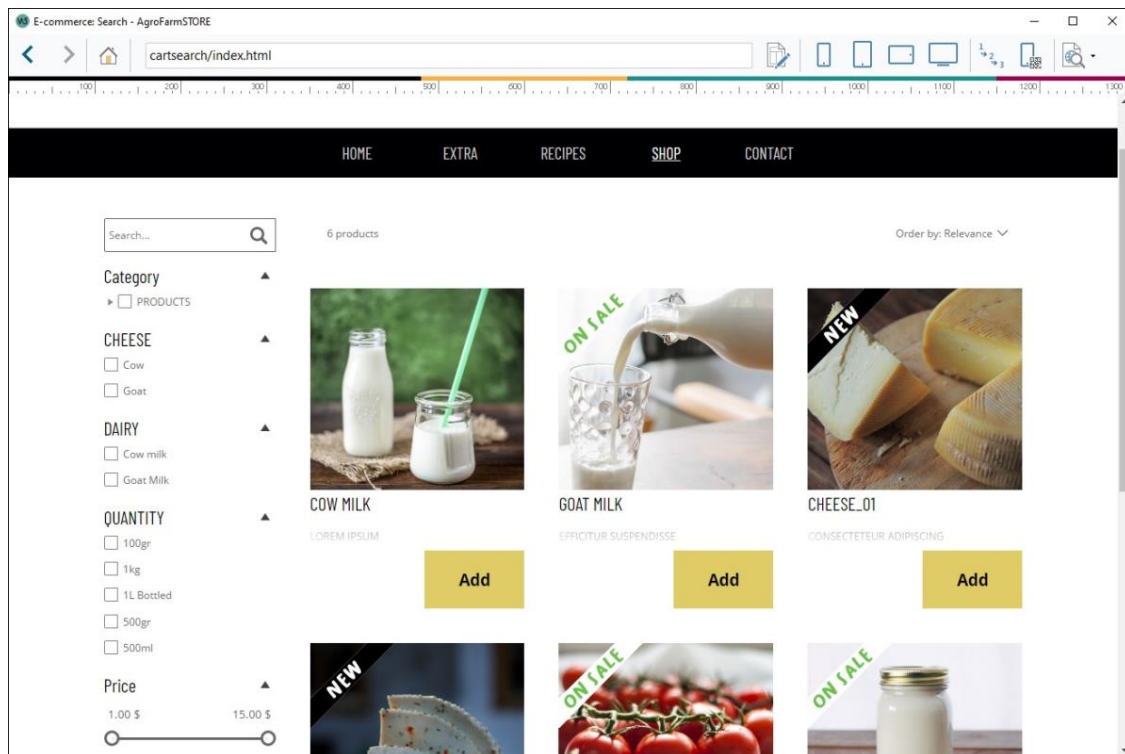


FIG:6.1.2

6.1.3 Cart and Order Placement

- Cart:
 - Real-time updates using state (in React) or localStorage for small apps.
 - Automatically updates subtotal.
- Checkout:
 - Collect delivery details in a secure form.
 - Simulate payment step (optional).
 - Clear cart and notify user of success.
 - Past Orders:
 - Displayed under the user's dashboard using user_id query to fetch data.

7. RESULTS

- **Authentication Works:** Users can sign up, log in, and stay logged in using JWT.
- **Dynamic Product Browsing:** Product list is fetched from the database with full filtering capabilities.
- **Real-Time Cart Update:** Items can be added, removed, and updated; subtotal updates accordingly.
- **Order Placement:** Upon checkout, an order record is created with full details.
- **Admin (Optional):** Admin can add new products, update stock, and delete listings using protected routes or a basic dashboard.

Optional testing tools:

- **Postman:** API test cases for login, product fetching, order creation.
- **Jest or Mocha:** For writing unit tests on backend logic.

8. CONCLUSION

The development of the Amazon Clone project has provided a comprehensive and practical insight into the domain of full-stack web development. It effectively bridges the gap between theoretical learning and real-world application by simulating a complete e-commerce platform. Through this project, core web technologies were utilized in a cohesive manner—HTML, CSS, JavaScript, and React.js on the frontend; Node.js with Express.js or Django on the backend; and MongoDB or MySQL for database management. Each layer of the stack was integrated to reflect real-world architectural patterns such as RESTful APIs, modular UI components, and secure data handling. From implementing authentication systems and CRUD operations to designing responsive user interfaces and deploying the application, every stage reinforced essential concepts required in the software development industry.

In summary, the project successfully achieves its objective of offering a real-time, hands-on learning platform while delivering a working prototype of an online marketplace. It lays a strong foundation for future advancements, such as integrating payment gateways, recommendation engines, or containerized deployments using Docker. The knowledge and experience gained through this project will greatly support continued learning and career growth in the field of web and software development.

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REFERENCES

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- Node.js Documentation: <https://nodejs.org/en/docs>
- MongoDB Docs: <https://mongodb.com/docs>
- W3Schools: <https://w3schools.com>
- GeeksforGeeks: <https://geeksforgeeks.org>
- Stack Overflow & GitHub Discussions

APPENDIX

A. SOURCE CODE

<https://github.com/OkoyeCharles/amazon-clone>